

CLASS & OBJECT

in Python



Student (Class)



("Alex", "Python", 2024)

Object 1



("Emma", "Data Science", 2023)

Object 2

What is a Class in Python?

A class is like a blueprint or template used to create objects.

It defines what data (attributes) an object will have and what actions (methods) it can perform.

Think of a Student

- A class represents the general idea of a student (name, course, year).
- An object is a real student created from that blueprint.



("Alex", "Python", 2024)
Object 1

("Emma", "Data Science", 2023)
Object 2

Defining a Python Class

We use the **class keyword** to define a class in Python.

Syntax



```
class ClassName:  
    # class definition
```

Example



```
class Student:  
    name = "Alex"  
    course = "Python"  
    year = 2024
```


Accessing Class Attributes

Class-level attributes can be accessed directly using the class name.

Example



```
class Student:
    name = "Alex"
    course = "Python"

print(Student.name)      # Output: Alex
print(Student.course)    # Output: Python
```

Explanation

- `Student.name` accesses data directly from the class.
- These attributes are shared by all objects unless overridden.

Class and Object in One Simple Example



```
class Student:
    name = "Emma"
    course = "Data Science"

    def show_details(self):
        print(f"{self.name} enrolled in {self.course}")

# Creating an object
student1 = Student()

print(student1.name)          # Output: Emma
student1.show_details()      # Output: Emma enrolled in Data Science
```

Key Points

- Student is the class.
- student1 is an object (instance).
- Dot notation is used to access attributes and methods.



Creating Multiple Objects from a Class

A single class can be used to create many objects with different data.



```
class Student:
    def __init__(self, name, course):
        self.name = name
        self.course = course

    def show_details(self):
        print(f"{self.name} → {self.course}")

student1 = Student("Alex", "Python")
student2 = Student("Emma", "AI")
student3 = Student("Liam", "Web Development")

student1.show_details() # Output: Alex → Python
student2.show_details() # Output: Emma → AI
student3.show_details() # Output: Liam → Web Development
```

Methods in a Python Class

A method is a function that belongs to a class and defines behavior.



```
class Student:
    def start_learning(self):
        print("Learning has started!")

student = Student()
student.start_learning()
```

Output



```
Learning has started!
```